Bounded-interface integral homology

Binary surface condensation, twisted attachments, and

provenance-aware terminals for compressed unknot hierarchies

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Abstract

The current ProveIt implementation already constructs a normal cut in binary cell families and extracts its all-dimensional source parallelity bundle, leaving a bounded inherited handle core. We develop the next algebraic layer: exact integral homology from that core and its bounded interface, without expanding the genera, boundary counts, or multiplicities of the attached surface bundles. The interacting part of H_1 is the cokernel of a matrix whose dimensions depend only on the core and the number of attached boundary circles. All other free ranks and repeated 2-torsion are binary counters. A saturated nonorientable base contributes one generator and one relation $2h + s = 0$, not one generator per crosscap. This yields a sharp mod-two torsion formula and a factor-two bound on torsion-exponent growth.

We also give bounded marked-port reconstruction, a torsion bound in terms of actual interface incidence, and compact cohomology witnesses. For genuine cut descendants of a certified knot exterior in S^3, a classical duality argument bypasses interior homology entirely: boundary genera determine all integral homology groups. Combined with the condensation theorem, this gives primitive-lattice and mod-two independence conditions on the attachments. A single sphere boundary is a ball terminal only under the embedding provenance, not for a generic torus-boundary input.

The accompanying standard-library implementation has independent algebraic replay, a checked small Smith kernel, and 4,800 seeded comparisons across six validation families. Paired complete homology-kernel measurements record both large-input gains and small-input regressions. These are interface-model measurements, not knot-recognition benchmarks. The source-geometric adapter, constructive cocycle lift, admissible patterned hierarchy, and global quasi-polynomial bound are not supplied by this contribution.

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1 Research position and the concrete bottleneck

1.1 The audited starting point

This continuation is based on ProveIt revision

`ed166080d7db14dd7a91d07788996f02947dd92a`.

The revision was committed on 12 October 2026. The relevant implementation is `Topology/UnknotRecognition/fast/`; the maintained theoretical account is under `synthesis/`. The source audit in Appendix A records specific paths and Git blob identifiers rather than treating the entire repository as a single reviewed object.

Two prior achievements are essential. The raw-cut constructor preserves all cell dimensions and source attaching charts in affine binary families [1]. More recently, `normal_cut_parallelity` extracts product handles in indices zero, one, and two, including cases where a base component has no selected zero- or one-handles. It retains base surface types, fibre twisting, and core-facing ports. Its inherited complementary core has at most $30T$ handles, with a conservative bound of $66T$ core ports, for T source tetrahedra [2]. These results are existing work, not contributions of the present article.

The inherited core is not yet an admissible patterned hierarchy structure. Likewise, a binary polygon description is not automatically a polynomial-time homology algorithm: an explicit chain matrix can have a number of rows proportional to the numerical normal weight. Our question is more specific: *which part of integral homology actually needs an explicit matrix once the core–bundle interface has been supplied?*

1.2 Relation to the published hierarchy algorithms

Polynomial-time extraction of parallelity data is not an absent theorem. Lackenby's Theorem 9.3 in *The efficient certification of knottedness and Thurston norm* computes the complementary core and the types and vertical attachments of parallelity components for an orientable normal surface, in time polynomial in the tetrahedron count and logarithmic normal weight [3]. That theorem and the current repository's all-dimensional source implementation are the proper antecedents of this work.

Lackenby's July 2026 manuscript gives an iteration bound $L(g+1)^L$, where L bounds hierarchy length and g bounds the relevant pattern complexity. It separately identifies the need to specify and bound the cost of individual operations [4, Section 9]. We address one such cost, not the entire search. The local algebra below cannot by itself bound the number of hierarchy corrections, the size of every later patterned core, or the cost of discovering the necessary surfaces.

1.3 What is new here, and what is classical

Mayer–Vietoris, the homology presentations of compact surfaces, Smith normal form, Alexander duality, and the Schoenflies theorem are classical [5, 6, 7]. The mathematical contribution here is their explicit bounded-interface organization: a signed integral matrix, a complete binary bulk formula, one additional generator per saturated twisted base, and interface conditions that are directly checkable downstream of source replay. We prove the resulting statements in full and implement the algebraic contracts. We do not claim that these consequences are previously unknown in every formulation in the literature.

The most immediately usable outcomes are the reduction in Theorem 3.1, the bounded-port extraction of Theorem 4.2, the twisted-attachment theorem 5.1, and the source-aware dispatch

rule in Section 7. The proofs remain useful even when a particular implementation chooses another normal-form backend.

2 The geometric and algebraic input contracts

2.1 A core with interval bundles

Let X be a compact orientable three-manifold, possibly disconnected, with nonempty boundary on every component. Assume a decomposition

$$X = K \cup \bigcup_i E_i, \qquad E_i \longrightarrow B_i, \qquad K \cap E_i = E_i|_{A_i}, \tag{1}$$

where K is a finite core, each E_i is an interval bundle over a compact connected surface B_i, and A_i is a compact one-dimensional submanifold of ∂B_i. Distinct bundle components are disjoint. Collars and corners are arranged so that these intersections give the usual Mayer–Vietoris sequence. In particular, there is no unrecorded triple intersection. A source certificate must justify this contract when the input comes from a normal cut.

Each component of A_i is either a closed interval or an entire boundary circle of B_i. Write a_i and c_i for their respective counts, and $b_i = |\pi_0(\partial B_i)|$. Adjacent core-facing arcs must first be merged through their actual quotient vertices. Counting raw port arcs as components of A_i is generally wrong.

For an orientable base let g_i be its genus. For a nonorientable base let $k_i \geq 1$ be its crosscap number. Thus

$$\chi(B_i) = 2 - 2g_i - b_i \quad \text{or} \quad 2 - k_i - b_i.$$

An orientable total interval bundle over a nonorientable base is twisted. The homology reduction uses the base's homotopy type; it does not infer bundle twisting from the slope of a singleton affine map.

Definition 2.1. A base with $b_i > 0$ is *saturated* if $c_i = b_i$: its entire boundary is attached through annuli. A saturated base has $a_i = 0$. Write $\mathcal{O}$ and $\mathcal{N}$ for the saturated orientable and nonorientable bases, and put $o = |\mathcal{O}|$, $t = |\mathcal{N}|$.

Unattached bundle components can be grouped in profiles with a binary multiplicity m_i. An attached component is individually identified and has $m_i = 1$. In particular, a port cannot silently refer to several identical copies. No loop in the intended computation is indexed by g_i, k_i, b_i, or m_i.

2.2 The core presentation and signed circle maps

Take a finite CW model for K, with v vertices, e edges, f two-cells, and u three-cells. A spanning forest in its one-skeleton gives

$$H_1(K;\mathbb{Z}) = \mathbb{Z}^d / \operatorname{im} R, \qquad d = e - v + b_0(K), \qquad R \in \mathbb{Z}^{d \times f}.$$

Every core two-cell has a closed edge word. Its column in R is the exponent sum on the nonforest edges. Collapsing the forest simply deletes tree-edge coordinates; it does not introduce long word substitutions.

For the attached boundary circles of B_i, let

$$z_{i1}, \ldots, z_{ic_i} \in \mathbb{Z}^d$$

be their image cycles in this presentation. For an orientable base, orient all its boundary components using one orientation of the base. This *relative* orientation information is indispensable. Local consistency around each individual circle does not establish consistency between different boundary circles of the same orientable base.

For a nonorientable base, use the boundary convention in the presentation

$$2(x_1 + \cdots + x_{k_i}) + d_1 + \cdots + d_{b_i} = 0. \tag{2}$$

Changing one boundary direction changes the eventual sum by $2z$ and is absorbed by a change of the retained generator. For saturated bases define

$$s_i = \sum_{j=1}^{b_i} z_{ij}. \tag{3}$$

A source implementation must provide these cycles, not just their number, a port mask, or an Euler characteristic.

2.3 The attachment graph

Let Γ have one vertex for every connected component of K and every attached base. Each component of A_i contributes one edge joining its base vertex to the component of K containing its image. Parallel edges are retained. Then

$$r_\Gamma = |E(\Gamma)| - |V(\Gamma)| + b_0(\Gamma). \tag{4}$$

Unattached bundles are not vertices of Γ; they are counted separately. The empty graph has all three counts zero. This convention handles an empty core without a special topological argument.

2.4 What the delivered input validator establishes

The Python model checks the integer domains, closed and contiguous core edge words, attachment counts, multiplicities, and graph incidence. These checks make a valid algebraic graph-of-spaces input. They do *not* establish that a native raw-cut packet realizes it, that K has the claimed three-manifold attaching maps, or that X embeds in S^3. Those are separate geometric premises. The homology formula for the graph of spaces is exact; its interpretation as the homology of a particular source cut requires source binding.

3 Exact integral condensation

3.1 The small matrix and binary bulk ranks

List the columns s_i for $i \in \mathcal{O}$ and $i \in \mathcal{N}$ as matrices S_O and S_N. Set

$$C = \begin{pmatrix} R & S_O & S_N \\ 0 & 0 & 2I_t \end{pmatrix} \in \mathbb{Z}^{(d+t)\times(f+o+t)}. \tag{5}$$

The extra row for a saturated nonorientable base corresponds to a generator h_i, and its column asserts $s_i + 2h_i = 0$.

Define the per-copy free contribution λ_i by Table 1. The cases $b_i = 0$ are necessarily unattached. The last column records only the isolated closed nonorientable contribution, not torsion that might be present in $\operatorname{coker} C$.

Base and attachment condition	λ_i	Extra $\mathbb{Z}/2$ per copy
Orientable, $b_i > c_i$	$2g_i + b_i - 1 - c_i$	0
Nonorientable, $b_i > c_i$	$k_i + b_i - 1 - c_i$	0
Orientable, $b_i = c_i > 0$	$2g_i$	0
Nonorientable, $b_i = c_i > 0$	$k_i - 1$	0
Orientable, $b_i = 0$	$2g_i$	0
Nonorientable, $b_i = 0$	$k_i - 1$	1

Table 1: Local surface data retained as binary counters.

Set

$$\beta = r_\Gamma + \sum_i m_i \lambda_i, \qquad \mu = \sum_{\substack{i:\ b_i=0 \\ B_i \text{ nonorientable}}} m_i. \tag{6}$$

The sum is over the supplied profiles; attached profiles have multiplicity one. These expressions involve ordinary integer arithmetic, not expanded lists of basis elements.

Theorem 3.1 (Bounded-interface integral homology)**.** *Under the contract of Section 2,*

$$H_0(X;\mathbb{Z}) \cong \mathbb{Z}^{b_0(\Gamma)+\sum_{A_i=\varnothing} m_i}, \tag{7}$$

$$H_1(X;\mathbb{Z}) \cong \operatorname{coker} C \ \oplus\ \mathbb{Z}^{\beta} \ \oplus\ (\mathbb{Z}/2)^{\mu}. \tag{8}$$

The inclusion of the core is represented by its original d generators in $\operatorname{coker} C$*. No matrix dimension in* (5) *depends on a numerical base genus, crosscap number, boundary count, or unattached multiplicity. The direct-sum splitting in* (8) *is not asserted to be canonical.*

Proof. The interval-bundle projection is a homotopy equivalence $E_i \simeq B_i$, and its restriction is a homotopy equivalence $E_i|_{A_i} \simeq A_i$. Mayer–Vietoris therefore gives the exact sequence

$$\bigoplus_i H_1(A_i) \longrightarrow H_1(K) \oplus \bigoplus_i H_1(B_i) \longrightarrow H_1(X) \longrightarrow \bigoplus_i H_0(A_i) \longrightarrow H_0(K) \oplus \bigoplus_i H_0(B_i). \tag{9}$$

An interval contributes zero to $H_1(A_i)$; an attached circle identifies its boundary generator with z_{ij}. The kernel of the final map is the integral cycle group of Γ, a free group of rank r_Γ. Thus the middle cokernel extends by $\mathbb{Z}^{r_\Gamma}$. An extension with free quotient splits as an abelian group. The same incidence map gives (7).

It remains to compute the middle cokernel explicitly. An orientable genus g base has the abelian presentation

$$\langle a_1, b_1, \dots, a_g, b_g, d_1, \dots, d_b \mid d_1 + \cdots + d_b = 0 \rangle .$$

When $b = 0$, the displayed relation is empty and its group is $\mathbb{Z}^{2g}$. For a nonorientable base, use (2). These presentations follow by abelianizing the standard polygon presentations; the commutator terms disappear in the orientable case [5].

Suppose first that $c < b$. Reorder boundary components so that the attached ones are $d_1, \dots, d_c$, and d_b is unattached. Eliminate d_b using the single surface relation. This is an integral elimination with coefficient one, including in the nonorientable case. Each attached relation then eliminates d_j in favor of z_{ij}. Since there is still the eliminated unattached boundary generator available, no relation is imposed on the core. The remaining free counts are $2g + b - 1 - c$ and $k + b - 1 - c$.

If $c = b > 0$ and the base is orientable, substitute all $d_j = z_{ij}$. The only residual relation is $s_i = 0$, while all $2g$ handle generators remain free. If the base is nonorientable, introduce

$$h = x_1 + \cdots + x_k, \qquad y_j = x_j \quad (2 \le j \le k).$$

This is a unimodular change of basis, since $x_1 = h - y_2 - \cdots - y_k$. The residual relation is $2h + s_i = 0$; the $k - 1$ generators y_j remain free. This accounts for exactly one new row and one new column in (5), irrespective of k.

Finally, a closed nonorientable base has $H_1 \cong \mathbb{Z}^{k-1} \oplus \mathbb{Z}/2$, by the same change of basis. An unattached bundle is a separate component of X, so each such copy contributes one independent $\mathbb{Z}/2$. An unattached orientable closed base contributes only $\mathbb{Z}^{2g}$. Applying these eliminations component by component gives (5), Table 1, and (8). All eliminations retain the original core generators, proving the inclusion assertion. □

3.2 Recovering the remaining Betti numbers

The decomposition gives

$$\chi(X) = v - e + f - u + \sum_i m_i \chi(B_i) - \sum_i a_i, \tag{10}$$

since a circle has Euler characteristic zero and an interval has Euler characteristic one. If $r = \operatorname{rank}_{\mathbb{Q}} C$, then

$$b_1(X) = d + t - r + \beta. \tag{11}$$

Corollary 3.2. *For an orientable three-manifold satisfying the nonempty-boundary hypothesis,*

$$H_3(X;\mathbb{Z}) = 0, \qquad H_2(X;\mathbb{Z}) \cong \mathbb{Z}^{\chi(X) - b_0(X) + b_1(X)}.$$

Proof. The top absolute homology vanishes on every component with boundary. Poincaré–Lefschetz duality identifies $H_2(X;\mathbb{Z})$ with $H^1(X, \partial X;\mathbb{Z})$. The latter is torsion-free: the possible universal coefficient term $\operatorname{Ext}(H_0(X, \partial X), \mathbb{Z})$ vanishes, and $\operatorname{Hom}(H_1(X, \partial X), \mathbb{Z})$ is free for this finite complex. Euler characteristic now gives the stated rank. □

This corollary must not be applied merely because an arbitrary algebraic model has an integer called `three_cells`. The delivered software reports the Euler value but does not certify the manifold premise or claim an unconditional H_2 computation from that count alone.

3.3 Two small examples with different behavior

Example 3.3 (A saturated annulus)**.** Let $H_1(K) = \mathbb{Z}\langle z \rangle$. Attach both boundary circles of an annulus to the same core component. The attachment graph has two parallel edges and $r_\Gamma = 1$. If the coherently oriented images are $z, -z$, then $C = (0)$ and $H_1(X) = \mathbb{Z}^2$. If they are z, z, then $C = (2)$ and $H_1(X) = \mathbb{Z} \oplus \mathbb{Z}/2$. The scalar data (χ, b, c) are identical.

Example 3.4 (A saturated Möbius base)**.** For a Möbius band with attached boundary image nz,

$$H_1(X) = \langle z, h \mid nz + 2h = 0 \rangle \cong \mathbb{Z} \oplus \mathbb{Z}/\gcd(n, 2).$$

Only the parity of n governs the torsion in this example. Replacing this base by a nonorientable base of k crosscaps adds $k - 1$ free generators, not a larger interacting matrix.

4 Recovering the bounded attachment interface

The repository already retains a bounded set of core-facing arcs, with source ranks and endpoint maps. The missing algebraic distinction is whether several such arcs form an interval or an entire boundary circle, and what that circle represents in the core. This section isolates a bounded postprocessing task rather than expanding all base polygons.

4.1 A single marked endpoint query

Suppose there are p raw core-facing arcs. Give the two endpoint occurrences of arc j distinct labels $2j$ and $2j+1$, even if their source corner points coincide. Assign endpoint occurrence r the weight 2^r. At a source point carrying several occurrences, add their weights. The quotient-vertex relation on the full base polygon system is already an interval-orbit problem in the existing geometry.

Lemma 4.1 (Endpoint masks do not alias)**.** *In the weighted quotient-vertex histogram, the nonzero orbit weights are exactly the bit masks of the endpoint equivalence classes. Each nonzero weight occurs on exactly one orbit. There are at most $2p$ nonzero weights, and every weight has at most $2p$ bits.*

Proof. Each endpoint occurrence is marked once. An orbit weight is consequently a sum of distinct powers of two. Its binary expansion records exactly which endpoint labels belong to that orbit; no carry or multiplicity ambiguity is possible. Distinct marked orbits have disjoint nonempty sets of labels, so cannot have the same nonzero weight. Every marked orbit contains at least one of the $2p$ endpoints. □

The unmarked orbits can be numerous; their multiplicity is irrelevant here. The output bound concerns this particular endpoint query, not an arbitrary weighted histogram with unrelated weights. A replayed orbit calculation is still needed to prove that the displayed masks are the correct geometric ones. Disjoint, exhaustive masks by themselves are not such a proof.

4.2 The finite port graph

Identify endpoint occurrences according to Lemma 4.1, and keep one edge for each raw core-facing arc. The resulting finite graph is exactly $A = \coprod_i A_i$. A loop contributes degree two at its vertex. Under the geometric contract, each vertex has degree one or two; every connected component is a path or a cycle. A path contributes one interval to a_i, whereas a cycle contributes one attached boundary circle to c_i.

Supply, for each port, the oriented edge path followed by its central line in a core graph. Endpoint images must agree after quotienting. Traversing a cycle concatenates these paths, reversing and inverting a path when the port is traversed backward. The result is a closed core word. For an orientable base, the traversal must follow the boundary direction certified by a single orientation of that base.

Theorem 4.2 (Bounded marked-port extraction)**.** *Given the correct quotient-vertex endpoint masks, coherent base-boundary orientations, and core paths of total word length P, the numbers a_i, c_i and all circle-image words can be constructed without visiting any unmarked base cell. The finite graph has at most $2p$ vertices and p edges. Each port and each letter of its core path is traversed a bounded number of times. The resulting interface and its verification have polynomial bit cost in $p + P$ and the bit length of the supplied labels and profiles.*

Proof. Decode the masks to a partition of $2p$ occurrences. Build the graph and check ownership of vertices, degree one or two, and matching core images. A traversal of each component distinguishes its two degree-one endpoints from the degree-two cycle case. Record one interval anchor for a path. For a cycle, concatenate its core paths in traversal order. Agreement at quotient endpoints proves contiguity and closure. Each edge occurs once in its component traversal, so the total emitted word length is at most P. The bit masks have the bounds in Lemma 4.1; sorting, graph traversal, and exact integer comparison are polynomial in the stated input size. No base genus or normal multiplicity is a traversal bound. □

The eight-port disc example. The maintained parallelity article describes a disc base supplied by an index-two product handle, with eight core-facing arcs and no selected index-zero or index-one handles [2]. Its attachment subset is a circle, so $a = 0$, $c = b = 1$, and $\chi(A) = 0$. Treating it as eight intervals would both alter Euler characteristic and lose its circle relation. The delivered eight-port test exercises this combinatorics, but is not a replay of that native triangulated example.

Exact scope of the implemented adapter. `ports.py` implements the mask decoder and the finite port graph construction. It checks local directions along each attachment component. It does not invoke the native weighted-orbit engine, recover source core paths, or certify relative orientations of different boundary circles. Those are the remaining inputs to a source adapter, not consequences of a port histogram or a successful algebraic replay.

5 Twisted attachments: torsion is tightly controlled

First perform the saturated orientable quotients and write

$$G = H_1(K;\mathbb{Z})/\langle s_i : i \in \mathcal{O}\rangle. \tag{12}$$

For the remaining saturated nonorientable bases, put

$$Q = (G \oplus \mathbb{Z}\langle h_1, \dots, h_t\rangle)/\langle s_j + 2h_j : 1 \le j \le t\rangle.$$

Thus $Q = \operatorname{coker} C$ in the notation of Section 3.

Theorem 5.1 (Twisted-attachment exact sequence)**.** *There is a natural short exact sequence*

$$0 \longrightarrow G \longrightarrow Q \longrightarrow (\mathbb{Z}/2)^t \longrightarrow 0. \tag{13}$$

Consequently:

1. *The odd-primary torsion subgroups of G and Q agree under inclusion.*
2. *If e is the exponent of* $\operatorname{Tor} G$*, with $e = 1$ when G is torsion-free, then the exponent of* $\operatorname{Tor} Q$ *divides $2e$.*
3. *If $G \cong \mathbb{Z}^r$, and $\overline{S}$ has columns $s_j \bmod 2$ in $G/2G$, then*

$$Q \cong \mathbb{Z}^r \oplus (\mathbb{Z}/2)^{\,t-\operatorname{rank}_{\mathbb{F}_2}\overline{S}}. \tag{14}$$

In particular, t twisted attachments do not produce a factor 2^t in the torsion exponent *bound.*

Proof. An element of G that vanishes in Q is an integer combination of the relations $s_j + 2h_j$. Comparing the independent h_j coordinates in $G \oplus \mathbb{Z}^t$ shows that every coefficient in this combination is zero. Hence $G \to Q$ is injective. Modulo G, the generators h_j have the independent relations $2h_j = 0$, proving (13).

An element of odd prime-power order in Q maps to zero in the elementary 2-group quotient, hence lies in G. Conversely, odd-primary torsion of G remains torsion of the same order by injectivity. This proves the first claim. For any torsion element $q \in Q$, the element $2q$ lies in G and is torsion there. Thus $2eq = 0$, proving the second claim.

When G is free, tensoring (13) with $\mathbb{Q}$ gives free rank r for Q, and the second claim says its torsion is elementary 2-torsion. The presentation has $r + t$ generators and t relations. Modulo two its relation matrix has rank $\operatorname{rank}_{\mathbb{F}_2} \overline{S}$, because the lower block $2I_t$ vanishes. Therefore

$$\dim_{\mathbb{F}_2}(Q/2Q) = r + t - \operatorname{rank}_{\mathbb{F}_2} \overline{S}.$$

Subtracting the contribution r from its free part gives (14). □

Example 5.2 (Large order is not large exponent)**.** Take $G = \mathbb{Z}/3\langle z \rangle$ and $s_j = z$ for all j. Then

$$Q \cong \mathbb{Z}/6 \oplus (\mathbb{Z}/2)^{t-1} \qquad (t \geq 1).$$

Its torsion order is $3 \cdot 2^t$, but its exponent is always 6. The test suite checks the five-attachment instance and 700 random free-core instances of (14).

5.1 Canonical torsion without expanding repeated factors

After a Smith computation on C, let its positive nonunit diagonal entries be $d_1 \mid \cdots \mid d_r$. The full torsion also contains μ copies of $\mathbb{Z}/2$, where μ may be a very large binary integer. Returning a list of μ entries would undo the compression.

Proposition 5.3 (Run-length invariant factors)**.** *The invariant factors of $\bigoplus_j \mathbb{Z}/d_j \oplus (\mathbb{Z}/2)^\mu$ can be represented by $O(r + 1)$ constant-value runs and computed in polynomial bit time in the input diagonal and* bits(μ)*. No integer factorization is required.*

Proof. Write $d_j = 2^{a_j} u_j$ with u_j odd. The a_j are nondecreasing and the u_j form a divisibility chain. Remove trivial odd factors. Merge μ additional exponent-one terms into the ordered positive a_j, retaining multiplicities in binary. Align this exponent sequence and the odd-factor sequence on the right, padding the shorter on the left with zero exponents or odd factor one. Multiply aligned factors.

For each prime, this is precisely the nondecreasing primary invariant-factor sequence padded on the left by trivial factors. Their termwise products are therefore the integral invariant factors. Only the 2-adic valuations are explicitly extracted; the odd factors remain whole integers. The exponent sequence has at most $r + 1$ runs and the odd sequence at most $r + 1$. Merging run endpoints uses a linear number of run operations and binary arithmetic. No run is expanded according to its multiplicity. □

6 Incidence bounds and normal-multiplicity safeguards

A small number of handles alone does not bound the bit size of their attaching maps. This distinction matters both for complexity and for mathematical statements about torsion.

Proposition 6.1 (Torsion from actual matrix incidence)**.** *For an integer matrix C, set $S = \sum_{ij} |C_{ij}|$. Then*

$$|\operatorname{Tor}(\operatorname{coker} C)| \le 3^{S/3}.$$

For the full compressed group in Theorem 3.1,

$$|\operatorname{Tor} H_1(X)| \le 2^{\mu} 3^{S/3}.$$

The latter is a mathematical bound, not a request to materialize 2^{μ}.

Proof. If the rank is zero, the first torsion group is trivial. Otherwise, let its rank be r. Smith theory identifies the torsion order with the greatest common divisor of the nonzero $r \times r$ minors. It is bounded above by any nonzero such minor. Hadamard's inequality bounds that determinant by the product of the Euclidean norms of its columns, and hence by their full column ℓ^1 norms $q_1, \dots, q_r$. Each q_j is a positive integer, $\sum q_j \le S$, and $q \le 3^{q/3}$ for every positive integer q. The last inequality is immediate for $q = 1, 2, 3$ and follows from the decrease of $(\log x)/x$ for $x \ge 3$. Multiplying proves the first bound. The isolated $\mathbb{Z}/2$ factors supply the second. □

Corollary 6.2 (Linear-incidence interfaces)**.** *Suppose a family of decompositions has total core two-cell word length and total attached-circle word length $O(T)$, and has $O(T)$ saturated twisted bases. Then the torsion order of every component meeting the core is $2^{O(T)}$, independently of all base genera and normal multiplicities. An unattached interval-bundle component has torsion order at most two.*

Proof. Forest collapse deletes edge coordinates. The total absolute mass of R and the boundary-sum columns is no larger than the total input word length. The lower block contributes $2t$. Thus $S = O(T)$. The matrix splits into blocks according to the connected components of Γ; an unattached bundle contributes no block and has the surface homology already computed. Apply Proposition 6.1 separately to each component. □

What this says about the native source charts. The current repository supplies $O(T)$ core and port records, with bounded total source-sector incidence [2]. To use Corollary 6.2 with that exact constant T, an adapter must verify that its core graph and port paths have *linear total word length*, not merely that their number of records is linear. The present package does not authenticate that refinement. A record containing a binary power of a long winding cannot be charged as a single letter. With general coefficient bit length B_C and q matrix rows, the safe alternative is the usual determinant estimate $\log |\operatorname{Tor} \operatorname{coker} C| = O(q(B_C + \log(q+1)))$.

6.1 Closed one-sided components cannot proliferate arbitrarily

Proposition 6.3. *If a compact orientable three-manifold M contains s pairwise disjoint closed connected nonorientable embedded surfaces, then $s \le b_1(M; \mathbb{F}_2)$. Consequently, the counter μ in an actual cut of a fixed source M is at most $b_1(M; \mathbb{F}_2)$.*

Proof. Each surface is one-sided. Choose an orientation-reversing loop on it. Push the loop slightly into its normal interval bundle: it starts on one local side and ends on the other. Closing it by a short fibre segment produces a loop that meets this surface once modulo two. The loop lies in that surface's bundle and is disjoint from all the other surfaces.

Intersection parity with each closed surface defines a class in $H^1(M; \mathbb{F}_2)$. Evaluating these classes on the constructed loops yields the identity matrix. The classes are linearly independent, proving the bound. For the second assertion, take the central closed base surfaces of the unattached nonorientable bundles. A cut is realized as a submanifold of M, so these surfaces are also pairwise disjoint in M. □

For a triangulation with T tetrahedra, the crude bound $b_1(M;\mathbb{F}_2) \le 6T$ follows from the number of source edges. Together with a verified linear-incidence adapter, this would bound the *total* torsion order over all cut components by $2^{O(T)}$. The very large multiplicity tests in our abstract model therefore must not be portrayed as normal cuts of a fixed small triangulation; they test the encoded algebraic contract beyond that geometric restriction.

7 A stronger shortcut for certified descendants in S^3

The generic API accepts a compact orientable triangulation with one torus boundary. This is a weaker promise than being the exterior of the input knot in S^3. The distinction permits a useful optimization only after source provenance is carried explicitly.

Theorem 7.1 (Boundary-determined integral homology in S^3)**.** *Let $X \subset S^3$ be a connected compact locally flat PL codimension-zero submanifold with nonempty boundary. Let b be the number of its boundary components and let G_∂ be their total genus. Then*

$$(H_0, H_1, H_2, H_3)(X;\mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^{G_\partial}, \mathbb{Z}^{b-1}, 0). \tag{15}$$

In particular, all these integral homology groups are torsion-free.

Proof. Let $\Sigma = \partial X$, a disjoint union of b closed orientable surfaces. Alexander duality gives

$$\widetilde{H}_0(S^3 - \Sigma;\mathbb{Z}) \cong H^2(\Sigma;\mathbb{Z}) \cong \mathbb{Z}^b.$$

Thus $S^3 - \Sigma$ has $b+1$ connected regions. The connected interior of X is exactly one region: it is open and closed in $S^3 - \Sigma$ and has no additional boundary there. Hence $S^3 - X$ has b components.

Let Y be the closure of $S^3 - X$. Its collar identifies Y up to homotopy with $S^3 - X$, and identifies $S^3 - Y = \operatorname{int} X$ up to homotopy with X. Apply Alexander duality to the compact locally contractible space Y. This gives

$$H_2(X;\mathbb{Z}) \cong \widetilde{H}^0(S^3 - X;\mathbb{Z}) \cong \mathbb{Z}^{b-1}, \qquad H_1(X;\mathbb{Z}) \cong H^1(S^3 - X;\mathbb{Z}).$$

The right-hand group in degree one is torsion-free by the universal coefficient theorem. It is finitely generated, since a collar makes the complement homotopy equivalent to a compact manifold. Finally, $H_3(X) = 0$ and doubling X along its boundary gives $2\chi(X) = \chi(\partial X) = 2b - 2G_\partial$. If $r = b_1(X)$, the already computed groups give $\chi(X) = 1 - r + (b-1) = b - r$. Therefore $r = G_\partial$. These duality statements are standard; see [5] and the related embedding obstructions in [8]. □

Corollary 7.2 (A source-aware sphere terminal)**.** *Under the hypotheses of Theorem 7.1, $b = 1$ and $G_\partial = 0$ imply that X is a three-ball. More generally, if $G_\partial = 0$, then X is homeomorphic to S^3 with the interiors of b disjoint three-balls removed.*

Proof. Each boundary component is a locally flat two-sphere. The Schoenflies theorem says that the closures of both complementary domains of such a sphere are three-balls [6, Theorem 5]. There are b complementary components of X, and b boundary components. Every complementary component meets at least one boundary component; thus each meets exactly one sphere and is a ball. Removing their interiors gives the assertion. For $b = 1$, either complementary closure is a ball. □

Why homology alone is not the terminal. The conclusion is a theorem about an embedded manifold, not an inference from a Betti vector. A punctured lens space has one sphere boundary and can have nontrivial torsion; a punctured integral homology sphere can have exactly the homology of a ball without being a ball. Neither is a counterexample to Corollary 7.2, because neither can satisfy its S^3 embedding premise in the nontrivial case. The existing generic sphere-boundary nonball control should therefore be retained, not removed.

7.1 Primitive relations and a parity obstruction

Corollary 7.3 (Necessary interface conditions in S^3)**.** *Assume the full decomposition* (1)*, including a regularized core K, lies in S^3. Then:*

1. *$\mu = 0$ and $H_1(K;\mathbb{Z})$ is free;*
2. *the subgroup generated by the saturated orientable sums s_i is a primitive sublattice of $H_1(K;\mathbb{Z})$;*
3. *writing G as in* (12)*, the saturated nonorientable sums are linearly independent in $G/2G$. In particular $t \le \operatorname{rank} G$.*

If X is connected, then

$$G_\partial = \operatorname{rank} G + \beta. \tag{16}$$

Proof. Theorem 7.1 applies to every connected component of X and K. Thus their first homologies are free. Formula (8) forces $\mu = 0$. Theorem 5.1 injects G into Q, and Q is a direct summand of the torsion-free group $H_1(X)$. Therefore G is torsion-free. A quotient of a free abelian group by a subgroup is torsion-free exactly when that subgroup is primitive, giving the second assertion.

Now G is free. Formula (14) and torsion-freeness of Q force $\operatorname{rank}_{\mathbb{F}_2} \overline{S} = t$. Rationally, $\operatorname{rank} Q = \operatorname{rank} G$; combine this with (8) and (15) for the last identity. □

These are necessary conditions, not an embedding recognizer. Passing a primitivity or parity test does not establish that the interface came from S^3. Conversely, torsion discovered under a claimed S^3 ancestry signals a defective ancestry, geometric adapter, or algebraic computation. It is not, by itself, a knottedness obstruction: nontrivial knot exteriors also embed in S^3.

7.2 How provenance can be preserved

An implementation can retain an embedding-ancestry DAG rather than expand coordinates in S^3 at every stage. Its root must bind a certified construction of the input knot exterior to the original diagram. Cutting out an open regular neighborhood of a properly embedded surface preserves the embedding by inclusion, and selecting components preserves it as well. A verified homeomorphic change of representation also preserves the existence of an embedding, by composition with the inverse homeomorphism.

An arbitrary quotient, filling, or gluing is not covered by these rules. Neither a hash nor a Boolean field named `certified` proves the root premise or a transition. This delivery includes no checker for such a DAG. The deliberately named `s3_boundary_consequences` computes only the consequent of Theorem 7.1; its returned record explicitly says that the embedding hypothesis has not been verified by that function.

Once the ancestry and the boundary profiles are independently replayed, (15) costs only binary arithmetic per profile. It can replace an interior matrix computation for Betti numbers or the predicate $b_1 = 0$. It does not supply the actual nonseparating surface, a boundary pattern, or the essentiality test needed by the hierarchy.

8 Compact cohomology witnesses

A hierarchy needs more than a nonzero Betti number. We can nevertheless separate finding an algebraic functional from lifting it to a geometric cutting surface.

Theorem 8.1 (A small functional in the condensed coordinates)**.** *Given the interface data, if $H^1(X;\mathbb{Z}) \neq 0$, then a nonzero integral functional on the presentation in Theorem 3.1 has a description polynomial in the encoded input size. It can be found with a polynomial-time integer normal-form algorithm. The description uses at most one binary-indexed bulk generator or an ordinary vector on the $d+t$ interacting generators. It need not enumerate a surface homology basis.*

Proof. If $\beta > 0$, choose a nonzero coordinate functional on one of the free summands recorded in (6). A directory consisting of the r_Γ graph generators followed by the surface-profile generators locates the chosen generator using a profile index, copy index, and local basis index, all in binary. If desired, choose the first nonempty family, which does not even require an index as large as the whole rank.

Otherwise, $H^1(X;\mathbb{Z})$ is the integer left kernel of C; finite torsion contributes no homomorphism to $\mathbb{Z}$. A polynomial-time integer normal-form algorithm gives a nonzero vector in this kernel whenever its rational nullity is positive [7]. Its coefficient bit lengths are polynomially bounded. Dividing the vector by the greatest common divisor of its entries makes it primitive without changing its support.

For completeness, the symbolic local coordinates can be interpreted in the standard surface presentations without expanding them. In an unsaturated base, eliminate one unattached boundary generator d_b and assign its functional value using the single surface relation. If one crosscap coordinate is set to one, only d_b needs the compensating value -2; if one free boundary generator is selected, only d_b needs -1. In a saturated nonorientable base use $x_1 = h - \sum_{j\geq 2} y_j$, $x_j = y_j$. A single y_j coordinate therefore requires only two nonzero original crosscap values. Orientable handle coordinates are already independent. A graph functional is represented on one nonforest edge of Γ. These are constant-support recipes with binary indices, or ordinary vectors of length $d+t$. □

The remaining geometric lift is real work. Theorem 8.1 does not identify an explicit source simplicial cocycle, let alone a normal surface. A native lift must transport the canonical base coordinates through the actual affine polygon identifications and respect the core inclusion. This requires certified chain maps or an equivalent source-level evaluation mechanism. In particular, reporting "$b_1 > 0$" and invoking a surface-construction theorem without supplying its required cochain is not an implementation.

A useful next target is a *port-preserving cocycle-lift theorem*: from a replayed affine base/core packet and one of the above functionals, construct a source-evaluable integral cocycle in polynomial encoded time. Its input and output must include the actual surface chart maps. This article isolates that question but does not claim its solution.

9 Bit complexity and its role in the quasi-polynomial goal

9.1 The local bound

Let N be the complete bit length of the supplied core, profile, and interface data, including all explicit attaching words. Let h be the number of core cells and p the number of raw ports.

The interface has at most p circle components and at most p interval components. In particular $o + t \le p$ and $d \le h$.

Theorem 9.1 (Polynomial encoded homology)**.** *From a supplied verified interface, constructing* (5)*,* (6)*,* (7)*, and* (10) *has polynomial bit cost in N. The matrix has $O(h + p)$ rows and columns. A polynomial-time integer Smith algorithm, followed by Proposition 5.3, computes a complete run-length description of $H_1(X; \mathbb{Z})$ in polynomial bit time. No dependence on the numerical genus or multiplicity is hidden in an output-length requirement.*

Proof. A spanning forest, core word checks, and attachment graph construction have polynomial cost. Every matrix coefficient is an exponent sum over supplied words, so its bit length is bounded by the word encoding size. There are at most $h + p$ generators and $h + p$ relations up to fixed constants. Dense assembly is therefore polynomial even without sparse optimization. The q profile contributions require $O(q)$ arithmetic operations on binary integers of polynomially bounded bit length. Products such as $m_i\lambda_i$ have bit length at most the sum of the factor bit lengths plus one.

Kannan–Bachem provides a polynomial integer normal-form algorithm, including controlled transformation matrices [7]. The small Smith output has polynomial size. The run-length merge in Proposition 5.3 retains binary multiplicities. Free summands are likewise represented by their rank. Hence the output is a compressed isomorphism type rather than an exponentially long basis. □

The theorem begins *after* source interface extraction. The actual repository's weighted discovery and replay costs must be added with their stated trace/partition accounting [2]. A polynomial postprocessor cannot retroactively prove a bound on a preceding discovery algorithm whose largest intermediate encoding has not been bounded.

9.2 The reference Smith kernel is not a Kannan–Bachem implementation

The delivered optional Smith kernel uses elementary Euclidean row and column operations, emitting a trace. It is exact and terminating. To see termination, at a pivot clear its column and row by division with remainder. Every nonzero remainder that becomes the new pivot has smaller absolute value. Once the pivot row and column are clear, an entry in the remaining block not divisible by the pivot is moved into the pivot row; another remainder again decreases the pivot. Eventually the pivot divides the remaining block and the algorithm advances to a smaller submatrix. Previously settled pivots remain isolated, and later integral operations preserve divisibility by them.

This argument is not a bound on all intermediate coefficient sizes. We do not claim that this particular implementation has Kannan–Bachem's worst-case polynomial bit complexity. The proved polynomial *algorithmic existence* result uses that published normal-form backend; the delivered prototype's normal-form cost is measured and replayed. The polynomial assembly stage can also be used independently of any Smith computation.

9.3 What would close the global accounting

Proposition 9.2 (A sufficient accounting criterion)**.** *Suppose a complete, correct hierarchy recognizer on an n-crossing diagram has at most $L(g + 1)^L$ iterations, each iteration costs at most $P(N)$ bit operations, and all preprocessing is bounded by the same total scale. If P is a fixed polynomial, $L = O(\log n)$, $g = n^{O(1)}$, and $\log N = O((\log n)^2)$, its total running time is $2^{O((\log n)^2)}$. More generally a sufficient bound for $2^{O((\log n)^3)}$ is*

$$\log P(N) + \log(L + 1) + L \log(g + 1) = O((\log n)^3).$$

Proof. Multiply the iteration bound by the largest per-iteration cost and take logarithms. Under the first hypotheses, both $\log P(N)$ and $L\log(g+1)$ are $O((\log n)^2)$; the extra logarithmic factor in L is smaller. The second assertion is the same calculation without specialization. □

This is sufficient accounting, not a proof that the actual hierarchy has these bounds. The present contribution removes numerical base genus and multiplicity from one matrix dimension. It does not establish bounded patterned-core encodings throughout every correction, a logarithmic depth, or the completeness and cost of the constructive hierarchy operations. The current complete recognizer's fallback is not replaced.

10 Implementation and verification artifacts

10.1 The module boundary

The package is an additive research component, intended for placement under `Topology/UnknotRecognition/research/interface_homology/`. No native recognition dispatch is modified. Its main modules are:

`model.py` and `condense.py`
: Strict integer validation, closed edge words, binary profiles, forest collapse, the attachment graph, the matrix C, and all bulk counters.

`ports.py` and `replay.py`
: Marked endpoint decoding and finite port assembly; independent replay of the algebraic condensation using graph traversal rather than the producer's disjoint-set implementation.

`smith.py` and `boundary.py`
: The small exact Smith kernel, elementary-operation replay, binary torsion run merging, a mod-two rank utility, and explicitly conditional boundary consequences for an S^3 descendant.

SymPy is a validation dependency only. The main package and benchmark kernel use the Python standard library. No third-party mathematical paper or font file is redistributed.

10.2 Certificate meanings

A condensation packet records the source-model SHA-256 fingerprint, chosen core forest, surviving edges, component labels, the matrix, saturated-base indices, graph counts, and bulk counters. Replay checks a spanning forest by independent component traversal, extracts exponent sums independently, and recomputes all formulas. The hash binds an algebraic model. It is not an assertion that the model came from the diagram.

A Smith trace allows only row/column swaps, adding an integer multiple of one row or column to a *different* row or column, and row negation. Every operation is unimodular. Replay applies the supplied operations and checks that the result is diagonal, has positive nonzero entries in a prefix, and satisfies successive divisibility. These checks prove equality of the presented abelian groups without relying on the pivot-selection code.

Operation and bit caps produce `WorkLimit`; they do not produce a partial homology group or a topological verdict. The replay operation cap limits the supplied trace. For untrusted deployment, external input-size and elapsed-time limits are still appropriate; the research API is not advertised as a hardened network service.

10.3 Binary serialization

Python environments can limit conversion of enormous integers to decimal strings. The delivered wire codec encodes integers as signed canonical hexadecimal strings under an explicit tag. It rejects ambiguous forms such as negative zero, leading zeroes, or a prefixed hexadecimal spelling inside that tag. Booleans remain booleans and are rejected where an integer is required. The fingerprint uses this codec, so values such as 2^{16384} do not require changing a process-wide decimal-conversion limit.

10.4 What the tests do and do not certify

The independent expanded oracle writes down the standard abelian surface presentations *before* the eliminations in Theorem 3.1. It uses its own breadth-first core forest. Small expanded matrices are compared using SymPy's Smith normal form [9]. A second family checks the reference Smith kernel against all determinantal divisors of small matrices. These are independent finite algebraic checks, not proofs of a source triangulation's geometry.

The current delivery has no Regina installation, does not execute the maintained repository suite, and does not replay the native cut-cell or parallelity certificates. It supplies neither a Lean formalization nor a full source-to-interface producer. These limitations are recorded in the machine-readable validation output as well as in the integration notes.

11 Completed experiments and reproducibility

11.1 Validation families

The delivered run passes all 25 unittest methods, with no failures, errors, or skips. Within those methods, six seeded families contain 4,800 cases:

Independent comparison or property	Cases
Expanded surface presentation versus condensation	1,200
Reference Smith diagonal versus SymPy	1,800
All determinantal divisors of small matrices	200
Binary repeated-2 invariant-factor merging	500
Free-core twisted-attachment formula	700
Oriented marked-port cycles	400

The six random seeds are the integers from `2026101201` through `2026101206`. Deterministic tests additionally cover empty matrices and cores, disconnected cores, interval cycles, both signs of boundary maps, Möbius attachments, closed surfaces, malformed models, forged packets, resource caps, and 16,385-bit parameters. The summary in `results/validation.json` records the environment and source hashes. These counts describe this delivery, not a rerun of any historical native checkpoint.

11.2 Paired complete homology-kernel measurements

Both timing arms start with the same abstract interface model and finish with a serialized H_0/H_1 answer. The condensed arm assembles and replays its packet, computes Smith form of the small matrix, replays that trace, and merges the binary factors. The expanded arm builds the literal standard surface presentation, runs the *same* Smith kernel, replays its trace, and serializes the

same invariant-factor answer. Thus the comparison isolates presentation compression rather than a change of normal-form library. The condensed arm includes its additional condensation replay.

There are eleven cases, one warmup round, and nine measured paired rounds. A third arm repeats the condensed computation as an A/A control. The three arms are shuffled each round with seed `2026101211`; garbage collection is performed outside each timer. All 330 warmup and measured samples finish, and all answers agree. Of these, 297 are measured samples. Tables report the median of the *paired* expanded/condensed ratios, not the ratio of two independently rounded medians. Values greater than one favor condensation.

Case	Condensed (ms)	Expanded (ms)	Paired ratio	A/A
$g = 0$	0.284	0.182	0.647	1.065
$g = 1$	0.290	0.199	0.677	1.038
$g = 8$	0.302	0.260	0.895	0.954
$g = 64$	0.299	0.777	2.507	0.995
$g = 512$	0.283	4.564	16.123	0.983
$g = 4096$	0.284	33.603	116.599	0.974
$m = 1$	0.235	0.141	0.625	1.044
$m = 8$	0.235	0.680	2.887	0.980
$m = 32$	0.239	24.992	108.174	1.013
core only tiny	0.215	0.113	0.517	0.980
core only relations	1.805	1.452	0.796	1.038

Table 2: Complete abstract H_1 kernels, including exact Smith replay and output. Nine measured rounds; medians and paired ratios. No source cut or knot-recognition time is included.

The saturated-genus family attaches an orientable genus-g, two-boundary base to a one-loop core with boundary images $z, -z$. Its compressed matrix is always 1×1, while the expanded presentation has $2g + 4$ rows and three columns; one row accounts for the attachment-graph cycle. The exact answer is $H_1 = \mathbb{Z}^{2g+2}$.

The unattached nonorientable family has m copies of a closed three-crosscap base. Its exact first homology is $\mathbb{Z}^{2m} \oplus (\mathbb{Z}/2)^m$. It is a legitimate abstract bundle model but, by Proposition 6.3, large m is not a cut of a fixed small source. The core-only controls have no bulk surface data and expose the extra packet/replay overhead.

The observed speedups are local. In particular, the genus-4096 case is a homology-model measurement, not a 4096-crossing knot or a measured native normal cut. Tiny cases regress, and even an excellent matrix-stage ratio says nothing about a source extractor that dominates the total runtime.

11.3 Large binary parameters without an expanded baseline

For $N = 2^e$, we also process N unattached copies of a closed nonorientable base with $N + 1$ crosscaps. The exact answer is

$$H_0 = \mathbb{Z}^N, \qquad H_1 = \mathbb{Z}^{N^2} \oplus (\mathbb{Z}/2)^N.$$

The interacting matrix is empty. Source-model hashing, packet replay, the homology result, and its serialization are included. The first of ten runs is treated as warmup.

The expanded arm is deliberately not run at these sizes, and no timing ratio against it is asserted. The purpose is to test that construction, replay, and output serialization really use bit length rather than numerical multiplicity. This is not evidence that a fixed-size normal triangulation realizes those particular large profiles.

e in $N = 2^e$	Matrix	Serialized answer bytes	Median ms
64	0×0	157	0.058
256	0×0	349	0.055
1,024	0×0	1,117	0.062
4,096	0×0	4,189	0.107
16,384	0×0	16,477	0.272

Table 3: Binary stress tests, with genus/crosscaps and multiplicity both large. No expanded baseline or timing ratio is claimed.

11.4 Reproduction commands

From the package root:

```
python validate.py
python benchmark.py
python make_tables.py
sh build.sh
```

The independent tests require SymPy; the implementation and benchmark kernel do not. The completed run used Python and SymPy versions recorded in the JSON files. Timing results are specific to this environment and should be regenerated under the project's normal native benchmark controls before selecting a production threshold.

12 Integration plan and corrections to the research interface

12.1 A narrow, testable native adapter

The proposed adapter should consume a successfully replayed `normal-cut-parallelity-v1` packet and produce four things: a core one-skeleton with two-cell words; the endpoint equivalence classes of all core ports; a central core path for each port; and relative boundary orientation data for orientable bases. Its source fingerprint must match the original triangulation and unscaled normal vector.

The existing base profiles already supply the genus or crosscap count, boundary count, multiplicity, and port ownership. They do not by themselves supply the ordered boundary cycles or their signed maps. The finite adapter in `ports.py` starts precisely after those source-level inputs have been obtained. It should not be connected to a guessed geometric map or a profile-only approximation.

Validation should proceed on small native cuts by comparing the resulting integral group with an independent expanded-cut homology calculation, including torsion, not just ranks over two finite fields. Relabelled source triangulations, one-sided multiples, high-degree edge links, and the index-two-only disc example should be included. Tests must also disable the producer while replaying saved source evidence, so that an algebraic agreement cannot mask identical geometry mistakes on both sides.

12.2 A separate provenance-aware boundary path

The S^3 optimization has a different adapter: a replayed diagram-to-exterior root, a cut/homeomorphism ancestry, and componentwise boundary profiles. Its conclusion is exact integral homology by Theorem 7.1; a single-sphere component additionally has a ball conclusion by

Corollary 7.2. The generic triangulation path remains available and retains the sphere-boundary nonball counterexample.

The algebraic path remains valuable even when the boundary shortcut applies: it can check necessary interface identities, retain core inclusion data, and prepare cohomology functionals. It is not necessary to compute Smith form merely to rediscover a torsion-free Betti vector already implied by verified S^3 provenance.

12.3 Documentary and specification corrections

The audited root README still uses the broad phrase "bounded-core geometry" in its earlier boundary-constructor checkpoint. The newer `parallelity.tex` proves a bounded *inherited* core. A precise status update should distinguish that achievement from the still-missing admissible patterned core; it should not continue to list all bounded-core construction as absent. Conversely, the statement that a bounded *admissible* core remains open is not contradicted by the inherited core result.

The generic raw-cut article correctly keeps interior homology separate from boundary data. That separation can be sharpened, not discarded: for inputs with certified S^3 ancestry, boundary data already determines all integral homology. For unrestricted torus-boundary triangulations, it does not. No existing nonball test should be reclassified as incorrect on the basis of the stronger, currently unverified source promise.

Finally, a source wrapper should avoid the phrase "polynomial homology implementation" without specifying its Smith backend and the cost of its geometric extractor. The mathematical polynomial-time theorem, the tested Euclidean prototype, and the native weighted trace bounds are three different assertions.

13 Further research questions

13.1 Source-bound port cycles and relative orientations

Can the existing affine source charts emit all core words and oriented attachment circles with a proved linear total incidence bound in T? The endpoint-mask theorem reduces the component-classification part to one marked vertex query and $O(p)$ finite graph data. The remaining issue is source-certified paths and the relative signs on all boundary circles of each orientable base. A useful first deliverable is an independently replayed adapter with mutation tests for whole-circle orientation reversals.

13.2 A verified S^3 ancestry discipline

Which current hierarchy operations preserve an explicit embedded descendant, and which preserve only an abstract homeomorphism type? A small proof kernel for root construction, cutting, component selection, and verified homeomorphisms would make the boundary-only homology and sphere terminal available without solving a new embedding problem at every stage. Operations that fill or identify boundary pieces require separately proved rules.

13.3 Source-evaluable cohomology and connected cutting surfaces

Can the symbolic functionals of Theorem 8.1 be pulled back through the affine bundle charts in polynomial bit time, with no expansion of g_i or m_i? The output should support evaluation on source edges and independent checking of the cocycle equation. A subsequent step must

produce the primitive nonseparating surface required by the hierarchy, with its own coordinate and construction bounds. Neither step follows from a rank count.

13.4 Marked inclusion data beyond H_1 as an abstract group

Can one retain, at bounded-interface cost, the maps from the individual boundary components into $H_1(X)$, intersection pairings, and the marked peripheral classes? These data are stronger than an isomorphism type and could guide a pattern-preserving simplification. The matrix C already retains the core inclusion, but its bulk direct-sum decomposition is not a canonical boundary marking.

13.5 Admissible patterned cores and repeated corrections

The inherited $30T$-handle core is an important local bound. How can it be converted to a uniform-type admissible structure while transporting every boundary pattern, and how does that encoding behave after a weak reduction or a discarded hierarchy suffix? A satisfactory theorem must bound the largest intermediate bit length, not just the number of handles after one normal cut.

13.6 Logarithmic depth and constructive balancing

A balance theorem must control both the topology and its construction cost. Can a Cheeger-region or multisurface argument bound hierarchy depth by $O(\log n)$ while keeping all correction interfaces within the same encoded budget? Proposition 9.2 identifies the accounting target, but no balanced-region search or depth proof is established here.

13.7 Fast exact algebra with portable evidence

The reference Smith trace is easy to replay but can exhibit coefficient growth. A production backend could emit bounded-size unimodular certificates from a polynomial normal-form algorithm, or use modular methods followed by exact reconstruction and replay. When the source is in S^3, the primitivity and parity constraints of Corollary 7.3 offer additional checks, but do not remove the need to authenticate the geometric source.

13.8 Native measurements rather than model extrapolation

The next benchmark should time the complete chain from a source triangulation and normal vector through geometric replay, interface construction, homology, and output. It should include the existing expanded cut oracle, repeated A/A controls, and cases where the direct boundary path is enabled only by certified ancestry. The small-input regressions in the present model measurements are a reason to measure dispatch policies, not to make condensation the unconditional default.

14 Conclusion

The interacting integral homology of a core with surface bundles is much smaller than the expanded surface homology suggests. Unsaturated bases contribute only free ranks; saturated orientable bases add one signed boundary-sum relation; saturated nonorientable bases add one relation $2h + s = 0$ and one generator. This yields a complete compressed homology presentation with independent algebraic replay and precise bit-cost separation from the source extractor.

For certified S^3 descendants, the stronger boundary-only route avoids even that matrix when only homology ranks or a sphere terminal are needed. Its premise must remain explicit. The primitive-lattice and parity consequences provide additional consistency checks, not a substitute embedding theorem. These local results advance the compressed hierarchy interface while leaving the construction, admissibility, depth, and global search bounds as explicit mathematical obligations.

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[3] Marc Lackenby, *The efficient certification of knottedness and Thurston norm*, arXiv:1604.00290v3, especially Section 9.4, Theorem 9.3. `https://arxiv.org/html/1604.00290v3`.

[4] Marc Lackenby, *Incompressible surfaces, hierarchies and unknot recognition*, arXiv:2607.23350v1, 25 July 2026, especially Sections 6, 9, and 10. `https://arxiv.org/abs/2607.23350`.

[5] Allen Hatcher, *Algebraic Topology*, Cambridge University Press, 2002. Surface homology, Mayer–Vietoris, universal coefficients, and duality. `https://pi.math.cornell.edu/~hatcher/AT/ATpage.html`.

[6] Morton Brown, *A proof of the generalized Schoenflies theorem*, Bulletin of the American Mathematical Society **66** (1960), 74–76, Theorem 5. `https://projecteuclid.org/journals/bulletin-of-the-american-mathematical-society/volume-66/issue-2/A-proof-of-the-generalized-Schoenflies-theorem/bams/1183523455.full`.

[7] Ravindran Kannan and Achim Bachem, *Polynomial algorithms for computing the Smith and Hermite normal forms of an integer matrix*, SIAM Journal on Computing **8** (1979), 499–507. `https://doi.org/10.1137/0208040`.

[8] Dmitry Tonkonog, *Embedding 3-manifolds with boundary into closed 3-manifolds*, Topology and its Applications **158** (2011), 1157–1162. `https://arxiv.org/abs/1003.3029`.

[9] SymPy developers, *Matrix normal forms*, version 1.14.0 documentation, accessed 12 October 2026. `https://docs.sympy.org/latest/modules/matrices/normalforms.html`.

A Source audit and proof-status inventory

A.1 Repository source pins

The audited commit is

`ed166080d7db14dd7a91d07788996f02947dd92a`.

The following paths are relative to `Topology/UnknotRecognition/`. These Git blob identifiers bind the inspected texts; they are not SHA-256 hashes of the newly delivered files.

`README.md`
`4fb929b5689d0bc20327e241cb775789660c8013`.

`synthesis/cut_cells.tex`
`dc12b1a58aedaed78f00b63109b0aa5214832ca1`.

`synthesis/parallelity.tex`
`9e6698d589568dafc35d7a39df7ad6f74ee46ca0`.

`fast/fastunknot/normal_parallelity.py`
`22cf15ffa11ca85e6a6c350f5d217f1649a01ea2`.

The initial part of `normal_cut_boundary.py` was also inspected, but its entire implementation was not audited. Code search for a Schoenflies reference returned no hit in the inspected repository; this is not an exhaustive novelty search. No claim is made to have read or rerun the many historical reports and archives.

A.2 Claim classes

Claim	Status in this delivery
Integral condensation and graph rank	Proved for the stated interface contract; standard-library implementation and independent algebraic replay.
Marked endpoint extraction	Proved and implemented downstream of the correct marked quotient and source paths; no native orbit invocation.
Twisted exact sequence and parity formula	Proved; 700 random free-core checks plus deterministic torsion tests.
Canonical binary torsion runs	Proved and implemented; 500 expanded comparisons and a 16,385-bit multiplicity case.
Polynomial-time homology theorem	Proved using a polynomial integer normal-form algorithm; the delivered Euclidean Smith kernel is not claimed to have that bit bound.
Boundary-only homology and sphere terminal	Classical consequences proved under certified S^3 embedding; the delivered helper does not verify the premise and is not a recognizer.
Source geometry and diagram ancestry	Adapter contracts and proposed proof rules only; not implemented or authenticated in this package.
Global quasi-polynomial recognition	Sufficient cost criterion and specific remaining questions; no complete algorithm or global bound claimed.

B Minimal executable example

The following example is an algebraic saturated Möbius attachment, not a knot input:

```
from interface_homology import Core, Loop, Surface, Model
from interface_homology import condense, verify_condensation
from interface_homology.smith import smith, verify_smith, torsion_blocks

core = Core(vertices=1, edges=((0, 0),))
```

```
base = Surface(orientable=False, complexity=1, boundary=1,
               circles=(Loop(anchor=0, word=(1, 1)),))
model = Model(core, (base,))
packet = condense(model)
assert verify_condensation(model, packet)
certificate = smith(packet.matrix, cols=packet.relation_count)
assert verify_smith(packet.matrix, certificate,
                    cols=packet.relation_count)
free_rank = packet.rows - len(certificate.diagonal) + packet.free_bulk
torsion = torsion_blocks(certificate.diagonal, packet.two_torsion_bulk)
assert free_rank == 1
assert torsion == ((2, 1),)  # Z plus one Z/2
```

The model-to-packet transition authenticates this algebraic input only. Changing the surrounding code comment to "a cut of the input knot" does not create the missing source evidence.

C Counterexamples that the interface must retain

C.1 An identical profile can hide arbitrary circle degree

Let the core be a solid torus. Attach a disc bundle along its annular vertical boundary, using a primitive slope on the core boundary whose image in $H_1(K) = \mathbb{Z}$ is p times a generator. This is the familiar solid-torus plus two-handle description of a punctured lens space when $p \geq 2$. Its boundary is a sphere, but the condensation gives

$$C = (p), \qquad H_1 = \mathbb{Z}/p.$$

The base type, number of ports, and Euler characteristic do not determine p. The actual core word does. This example also shows why handle count alone is not a coefficient-size bound. It is not asserted to arise from normal cuts with one fixed small source triangulation.

C.2 Intervals can create homology with no attached circles

Take two contractible core components and a disc base attached by three intervals, two to the first component and one to the second. There are three graph edges, three graph vertices, and one connected component, so $r_\Gamma = 1$. The base and core have no first homology, and C is empty, but $H_1(X) = \mathbb{Z}$. Omitting the attachment graph would return zero.

C.3 Genus-zero boundary with several components is not a ball

The product $S^2 \times I \subset S^3$ has two sphere boundary components. It has $H_1 = 0$ but $H_2 = \mathbb{Z}$ and is not a ball. The boundary shortcut must retain the number of boundary components, not just their total genus.

C.4 A homology result is not a marked pattern

A solid torus and a nontrivial knot exterior both have boundary genus one and integral homology $(\mathbb{Z}, \mathbb{Z}, 0, 0)$. Thus the boundary shortcut does not recognize an unknot from its initial exterior. The further hierarchy and peripheral/pattern information remain necessary even when every homology query is cheap.